5G/6G COMMUNICATION ECA-5605

DESIGN TASK

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SLOT-**A**

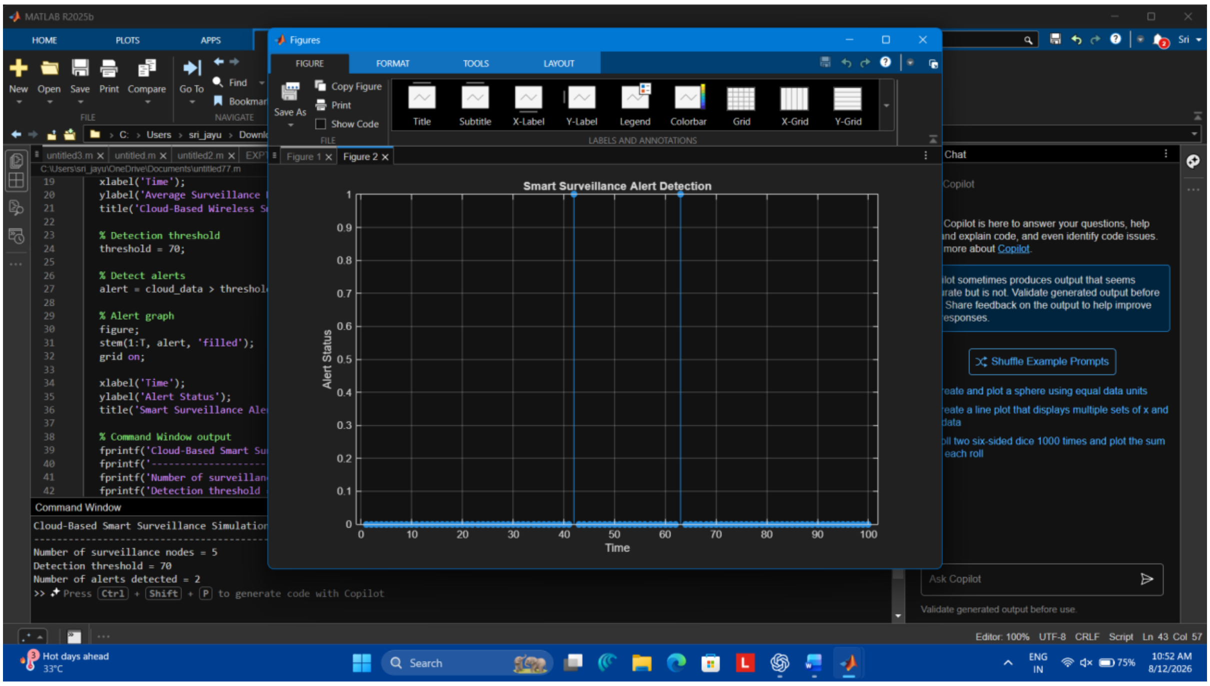
1. Cloud-Based Wireless Framework for Smart Surveillance

Aim:
To propose and simulate a cloud-based wireless framework for a smart surveillance system in MATLAB, where camera/sensor data is transmitted wirelessly to a cloud server for monitoring and analysis.

MATLAB Code:

```
clc;

clear;

close all;


% Number of surveillance nodes

N = 5;


% Simulation time

T = 100;


% Generate sensor/video data

data = randi([0 100], N, T);


% Wireless transmission

transmitted_data = data;


% Cloud processing

cloud_data = mean(transmitted_data, 1);
```

```
% Display results
figure;
plot(1:T, cloud_data, 'LineWidth', 2);
grid on;
xlabel('Time');
ylabel('Average Surveillance Data');
title('Cloud-Based Wireless Smart Surveillance');

% Detection threshold
threshold = 70;
alert = cloud_data > threshold;

figure;
stem(1:T, alert, 'filled');
grid on;
xlabel('Time');
ylabel('Alert Status');
title('Smart Surveillance Alert Detection');

fprintf('Cloud-based surveillance simulation completed.\n');
fprintf('Number of wireless nodes: %d\n', N);
fprintf('Detection threshold: %d\n', threshold);
fprintf('Number of alerts detected: %d\n', sum(alert));
```



Result:

The MATLAB simulation successfully models a cloud-based wireless smart surveillance framework. Data from multiple surveillance nodes is transmitted to the cloud, where it is processed and analyzed. When the average surveillance data exceeds the predefined threshold, an alert is generated. The graphs show the received cloud data and detected surveillance alerts.

2. LEO Satellite-Assisted Communication Model

Aim:

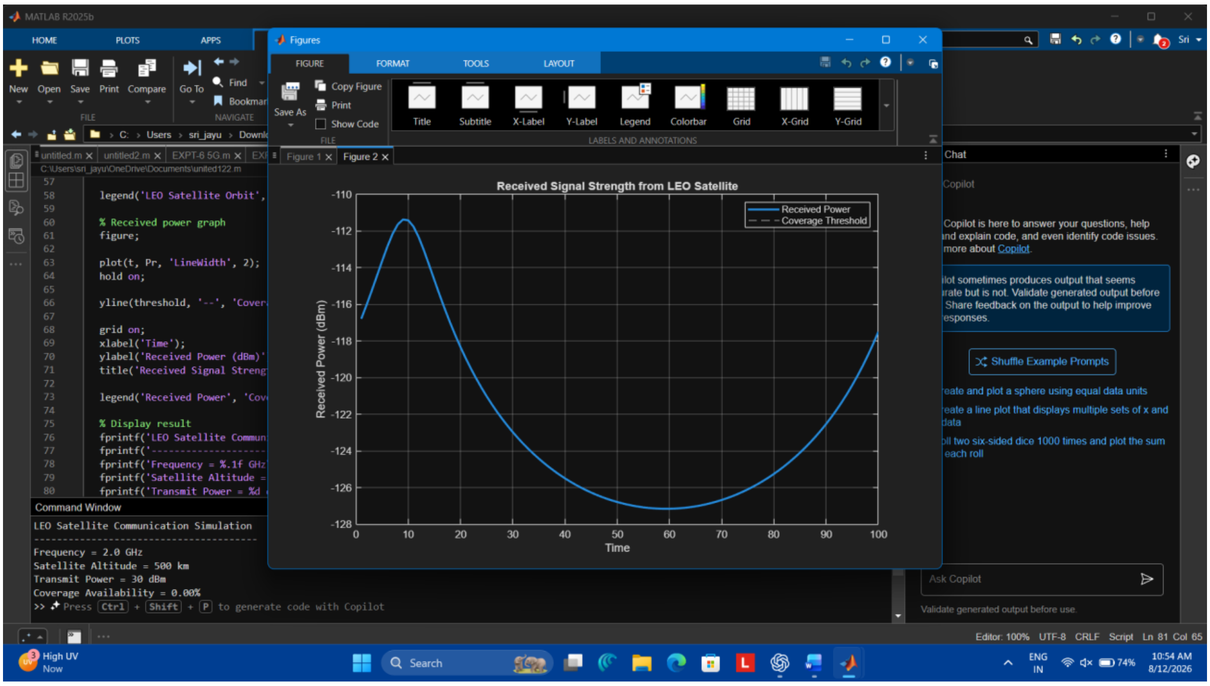
To design and simulate a satellite-assisted wireless communication model using a Low Earth Orbit (LEO) satellite to provide communication coverage to remote areas.

MATLAB Code:

```
clc;

clear;

close all;


% Simulation parameters

T = 100;

t = 1:T;
```

```
% Ground station and remote user positions
ground_station = [0 0];
remote_user = [80 20];

% LEO satellite orbit
sat_x = 100 * cos(2*pi*t/T);
sat_y = 100 * sin(2*pi*t/T);

% Distance between satellite and remote user
distance = sqrt((sat_x - remote_user(1)).^2 + ...
            (sat_y - remote_user(2)).^2);

% Free-space path loss model
frequency = 2e9;       % 2 GHz
c = 3e8;               % Speed of light
lambda = c/frequency;

path_loss = 20*log10(4*pi*distance/lambda);

% Received signal strength
transmit_power = 30;   % dBm
received_power = transmit_power - path_loss;

% Communication coverage threshold
threshold = -110;
coverage = received_power > threshold;

% Plot LEO satellite trajectory
figure;
```

```
plot(sat_x, sat_y, 'LineWidth', 2);
hold on;
plot(ground_station(1), ground_station(2), 'o', 'MarkerSize', 8);
plot(remote_user(1), remote_user(2), 's', 'MarkerSize', 8);
grid on;
xlabel('X Position (km)');
ylabel('Y Position (km)');
title('LEO Satellite-Assisted Communication Model');
legend('LEO Satellite Orbit','Ground Station','Remote User');

% Plot received signal strength
figure;
plot(t, received_power, 'LineWidth', 2);
hold on;
yline(threshold, '--');
grid on;
xlabel('Time');
ylabel('Received Power (dBm)');
title('LEO Satellite Received Signal Strength');
legend('Received Power','Coverage Threshold');

fprintf('LEO satellite communication simulation completed.\n');
fprintf('Coverage availability: %.2f%%\n', mean(coverage)*100);
```



Result:

The MATLAB simulation successfully models a LEO satellite-assisted communication system for remote wireless coverage. The satellite orbit and communication link between the satellite and remote user are represented graphically. Received signal strength is calculated using the free-space path-loss model, and the coverage periods are identified based on the specified signal-strength threshold.